

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

GridBalance — Frontend UI Failure Diagnosis & Repair Audit

1. Executive Summary

This diagnosis identifies the exact technical root causes behind the broken frontend UI layout and establishes the step-by-step repair strategy to restore the GridBalance visual design system.

2. Root Cause Analysis

Component / File	Identified Defect	Impact on Application UI
src/index.css	Missing Tailwind Import (@import "tailwindcss";) and residual Vite template rules (#root { width: 1126px; margin: 0 auto; text-align: center; }, h1 { font-size: 56px; }).	All Tailwind utility classes were ignored. The entire application was forced into a 1126px centered box with text-align: center, default unstyled buttons, huge 56px headings, and broken layouts.
postcss.config.js	0-byte empty file. @tailwindcss/postcss plugin was not registered.	Vite's CSS pipeline could not process Tailwind CSS v4 directives.
index.html	Default frontend header title.	Unprofessional browser tab label.
Typography Hierarchy	Heading rules in index.css overriding inline Tailwind font sizes.	Section headers (METER TAMPERING DETECTION) appeared disproportionately massive and collided with surrounding text.
Responsive Layout	Root container enforcing fixed width instead of full-width flex/grid layout (w-full min-h-screen bg-slate-950).	Layout collapsed into a single centered column on all screen sizes.

3. Repair Strategy

1. Configure PostCSS & Tailwind v4:

- Update postcss.config.js to register @tailwindcss/postcss.
- Update src/index.css to import @import "tailwindcss"; and replace Vite template CSS overrides with clean resets.

2. Design System & Palette Alignment:

- Dark technical command-center palette:

- Background: #090B10 / #0D1117 (bg-slate-950)
- Surface Cards: #11151D / #151A23 (bg-slate-900, border-slate-800)
- Primary Accents: Cyan (#22d3ee), Emerald Green (#34d399), Amber (#fbbf24), Rose (#f87171)
- Typography: Clean sans-serif headers (24–32px page titles, 16–18px section headers), monospace metrics (12–14px).

3. Application Shell & Layout:

- Full-width dark background (min-h-screen bg-slate-950 text-slate-100 flex flex-col).
- Sticky navbar header with live backend status badge (● ML Engine Online / ● ML Engine Offline).
- Main content container with max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 w-full flex-1.
- Footer with research evaluation disclaimers and Swagger API documentation links.

4. Component Refactoring:

- Refactor Overview Dashboard (/), Detection (/detection), Batch Detection (/detection/batch), Meter Profile (/detection/:meterId), Load Forecasting (/forecast), Model Intelligence (/models), and Research (/research) to enforce card layouts, status strips, input form styling, Recharts visualization, and responsive grids.